6.4 Multiplying and dividing: activities

← Unit overview

1. Groups of things

You need: €10 notes and €1 coins (real or play), small bags or cups.

1. "Put €23 in each of 4 cups." He builds it: 2 tens and 3 ones in each cup.
2. "How much is that altogether?" Gather all the tens (8 tens = 80) and all the ones (12). 80 + 12 = 92. That's 4 × 23 by splitting.
3. Now reverse it: "Here is €92. Share it fairly between 4 cups." Share the tens first (2 each, 1 ten left over), swap the leftover ten for ten €1 coins, then share the ones. That's 92 : 4 = 23.

Talk about it: "Why did we have to swap a ten?"

2. Area model on squared paper

You need: squared paper, coloured pencils.

Draw a rectangle 4 squares high and 67 squares wide (or let one square be 1 and just draw it roughly, labelling the sides). Draw a line to split it into a 4 × 60 part and a 4 × 7 part. Colour each part differently and write its area inside.

```
        60            7
   +------------+------+
 4 |    240     |  28  |     240 + 28 = 268
   +------------+------+
```

Try 6 × 284 split into three parts: 200, 80 and 4.

Talk about it: "Where are the two parts of the sum in the picture?"

3. Party planner

You need: a supermarket folder or website, paper.

Plan a party for his class (or a family gathering). How many children? Each child gets 2 drinks, 3 crisps bags, 1 cupcake. Drinks come in packs of 6, crisps in packs of 8. How many packs? What does it cost?

This mixes multiplying (how many altogether) and dividing with a remainder (how many packs).

Talk about it: "Did you round the number of packs up or down? Why?"

4. Sharing out

You need: a bag of dried pasta, raisins or small sweets (count them first; around 50–100).

Count the bag together. "If we share these between 3 people, how many each? Any left over?" Estimate first, then share and check. Try 4, 5, 6 people.

Talk about it: "Which number of people gave no leftovers? Why?"

5. Remainder stories

You need: paper.

You give a division with a remainder, e.g. 30 : 4 = 7 rest 2. He makes up three short stories:

- one where the answer is **7** (round down): “30 euros, cinema tickets cost 4 euros, how many tickets?”
- one where the answer is **8** (round up): “30 children, 4 in a car, how many cars?”
- one where the leftover matters: “30 sweets for 4 children, how many each and how many for Mum?”

Talk about it: “Why can the same sum have different answers in real life?”

Conversation prompts (any time)

- “Which is more: 30 × 40 or 40 × 30? Do you need to calculate?”
- “If 6 × 25 = 150, what’s 12 × 25?”
- “Is 99 × 4 easy or hard? What could you do?” (100 × 4 − 4.)
- “When did you last share something fairly? Was there a remainder?”